Integration by Parts

ANSWER KEY

The integration by parts formula

1 Evaluate $\int xe^x\,dx$ using integration by parts.

Let $u = x$, $dv = e^x\,dx$: $= xe^x - \int e^x\,dx = xe^x - e^x + C$.

2 Evaluate $\int x\cos x\,dx$ using integration by parts.

Let $u = x$, $dv = \cos x\,dx$: $= x\sin x - \int \sin x\,dx = x\sin x + \cos x + C$.

3 Evaluate $\int \ln x\,dx$ using integration by parts (with $dv = dx$).

Let $u = \ln x$, $dv = dx$: $= x\ln x - \int x \cdot \frac{1}{x}\,dx = x\ln x - x + C$.

Repeated and definite applications

4 Evaluate $\int x^2 e^x\,dx$, applying integration by parts twice.

First pass ($u = x^2$): $x^2e^x - 2\int xe^x\,dx$. Second pass gives $\int xe^x\,dx = xe^x - e^x$. Total: $x^2e^x - 2xe^x + 2e^x + C$.

5 Evaluate $\int_0^1 xe^{2x}\,dx$ exactly.

Let $u = x$, $dv = e^{2x}dx$: $\left[\frac{x}{2}e^{2x} - \frac{1}{4}e^{2x}\right]_0^1 = \left(\frac{e^2}{2} - \frac{e^2}{4}\right) - \left(0 - \frac{1}{4}\right) = \frac{e^2}{4} + \frac{1}{4}$.

Choosing u and dv wisely (LIATE)

6 Evaluate $\int x\ln x\,dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as u under the LIATE rule).

$u = \ln x$, $dv = x\,dx$, so $du = \frac{1}{x}dx$, $v = \frac{x^2}{2}$: $= \frac{x^2}{2}\ln x - \int \frac{x^2}{2} \cdot \frac{1}{x}\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$.

7 Evaluate $\int x^2\cos x\,dx$, applying integration by parts twice.

First: $u = x^2$, $dv = \cos x\,dx$: $x^2\sin x - 2\int x\sin x\,dx$. Second (on $\int x\sin x\,dx$): $= -x\cos x + \sin x$. Total: $x^2\sin x + 2x\cos x - 2\sin x + C$.

The recurring-integral trick

8 Evaluate $\int e^x\sin x\,dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.

Let $I = \int e^x\sin x\,dx$. First pass ($u = \sin x$, $dv = e^x dx$): $I = e^x\sin x - \int e^x\cos x\,dx$. Second pass on the new integral ($u = \cos x$): $\int e^x\cos x\,dx = e^x\cos x + \int e^x\sin x\,dx = e^x\cos x + I$. So $I = e^x\sin x - e^x\cos x - I$, giving $2I = e^x(\sin x - \cos x)$, so $I = \frac{e^x(\sin x - \cos x)}{2} + C$.

Definite integrals by parts

9 Evaluate $\int_0^{\pi} x\sin x\,dx$ exactly.

$u = x$, $dv = \sin x\,dx$: $[-x\cos x + \sin x]_0^{\pi} = (-\pi\cos\pi + \sin\pi) - (0) = \pi$.

10 Evaluate $\int_1^{e} x^2 \ln x\,dx$ exactly.

$u = \ln x$, $dv = x^2 dx$: $\left[\frac{x^3}{3}\ln x - \frac{x^3}{9}\right]_1^e = \left(\frac{e^3}{3} - \frac{e^3}{9}\right) - \left(0 - \frac{1}{9}\right) = \frac{2e^3}{9} + \frac{1}{9} = \frac{2e^3+1}{9}$.

Reduction formulas

11 Use integration by parts with $u = x^n$, $dv = e^x dx$ to derive the reduction formula $\int x^n e^x\,dx = x^n e^x - n\int x^{n-1} e^x\,dx$.

$du = nx^{n-1}dx$, $v = e^x$: $\int x^n e^x\,dx = x^n e^x - \int e^x \cdot nx^{n-1}\,dx = x^n e^x - n\int x^{n-1} e^x\,dx$.

12 Use the reduction formula above (applied twice) to evaluate $\int x^2 e^x\,dx$, confirming it matches an earlier result.

$\int x^2 e^x\,dx = x^2 e^x - 2\int x e^x\,dx$. Applying again: $\int x e^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x$. Total: $x^2 e^x - 2(xe^x - e^x) = x^2 e^x - 2xe^x + 2e^x + C$, matching the earlier double-integration-by-parts result.